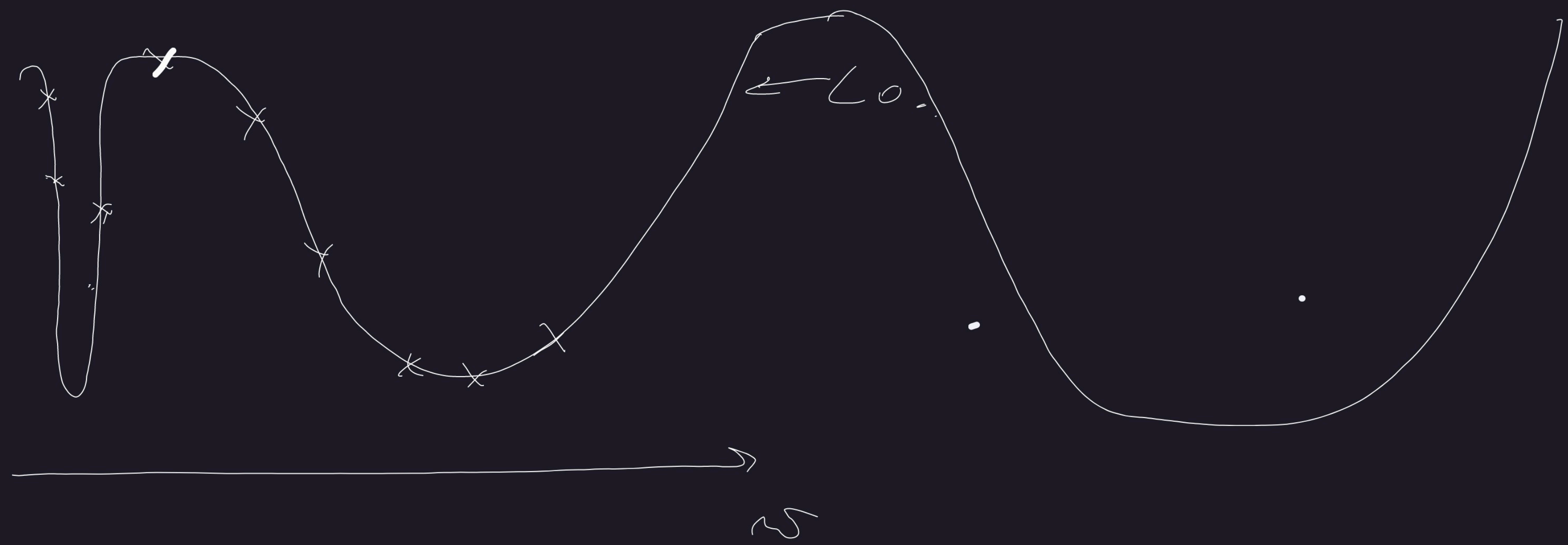
$$F(w) = \frac{1}{N} \sum_{i=1}^{N} f_i(w) \to \min_w$$

SGD:

$$\begin{cases} I_k \subset \mathrm{Unif}(1, 2, \ldots, N) \\ g_k = \dfrac{1}{|I_k|} \sum\limits_{i \in I_k} \nabla f_i(w_k) \\ w_{k+1} = w_k - \alpha_k g_k = w_k - \alpha_k \left(\nabla F(w_k) + \varepsilon_k \right) \end{cases}$$

$$\Sigma_k \xrightarrow[|I_k| \to N]{} 0$$

$$\varepsilon_k \sim \mathcal{N}(0, \Sigma_k)$$



LO
ω

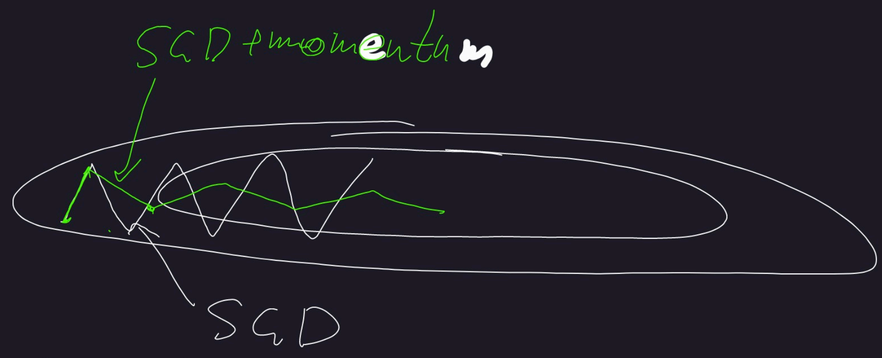
$$F(w) \to \min_{w}$$

GD: $w_{k+1} = w_k - \alpha_k \nabla F(w_k)$

Newton: $w_{k+1} = w_k - \alpha_k \left(\nabla^2 F(w_k)\right)^{-1} \nabla F(w_k)$

L-BFGS: $w_{k+1} = w_k - \alpha_k \, \text{L-BFGS}\left(\nabla F(w_k), \nabla F(w_{k-1}), \dots, \nabla F(w_{k-m}), w_k, \dots, w_{k-m}\right)$

SGD + momentum



$$w_{k+1} = w_k - \alpha_k m_k$$

$$m_k = \beta m_{k-1} + g_k \,, \quad 0 < \beta < 1$$

$$m_0 = g_0$$

$$w_1 = w_0 - \alpha_0 g_0$$

$$m_1 = \beta g_0 + g_1 \,, \quad w_2 = w_1 - \alpha(\beta g_0 + g_1)$$

$$m_2 = \beta^2 g_0 + \beta g_1 \,, \quad w_3 = w_2 - \alpha(\beta^2 g_0 + \beta g_1 + g_2)$$

$$\ldots$$

$$w_{k+1} = w_k - \alpha \sum_{i=0}^{k} \beta^{k-i} g_i$$

$$F(w) = \frac{1}{N}\sum_{i=1}^{N} f_i(w) + \frac{\lambda}{2}\|w\|_2^2 \to \min_w$$

SGDW: $$w_{k+1} = w_k - \alpha_k (m_k + \lambda w_k)$$

RMSprop

$$w_{k+1,i} = w_{k,i} - \alpha_k \left(\frac{g_{k,i}}{\sqrt{\delta_{k,i} + \varepsilon}} + \lambda w_{k,i} \right)$$

$$\delta_{k,i} = (1-\beta)\,\delta_{k-1,i} + \beta\, g_{k,i}^2$$

ADAMW

$$\omega_{k+1,i} = \omega_{k,i} - \alpha_k \left(\frac{m_{k,i}}{\sqrt{\delta_{k,i}+\varepsilon}} + \lambda \omega_{k,i} \right)$$

$$\hat{m}_{k,i} = (1-\beta_1)\hat{m}_{k-1,i} + \beta_1 g_{k,i}$$

$$\hat{\delta}_{k,i} = (1-\beta_2)\hat{\delta}_{k-1,i} + \beta_2 g_{k,i}^2$$

$$m_{k,i} = \frac{\hat{m}_{k,i}}{1-(1-\beta_1)^k}$$

$$\beta \sum_{i=1}^{k} (1-\beta)^{k-i} = \beta\left((1-\beta)^{k-1} + (1-\beta)^{k-2} + \dots + 1-\beta+1\right) = \cancel{\beta}\,\frac{1-(1-\beta)^k}{\cancel{1-(1-\beta)}} = 1-(1-\beta)^k$$

$$m_1 = \beta g_1$$

$$m_2 = (1-\beta)\beta g_1 + \beta g_2$$

$$m_3 = (1-\beta)^2\beta g_1 + (1-\beta)\beta g_2 + \beta g_3$$

$$m_k = \beta \sum_{i=1}^{k} (1-\beta)^{k-i} g_i$$

$$\mathbb{E} m_k = \beta \sum_{i=1}^{k} (1-\beta)^{k-i} \mathbb{E} g_i$$

α_k: ADAM $\alpha_k = 3 \cdot 10^{-4}$

SGD $\alpha_k = 10^{-3}, 10^{-2}$

$$\begin{bmatrix} x_1 & x_2 & x_3 \\ x_4 & x_5 & x_6 \\ x_7 & x_8 & x_9 \end{bmatrix} * \begin{bmatrix} k_1 & k_2 \\ k_3 & k_4 \end{bmatrix} \longrightarrow \begin{bmatrix} y_1 & y_2 \\ y_3 & y_4 \end{bmatrix}$$

$$y_{ij} = \sum_{p,q=0}^{w-1,\,h-1} k_{pq}\, x_{i+p,\,j+q}$$

$$\begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{pmatrix} = \underbrace{\begin{pmatrix} k_1 & k_2 & 0 & k_3 & k_4 & 0 & 0 & 0 & 0 \\ 0 & k_1 & k_2 & 0 & k_3 & k_4 & 0 & 0 & 0 \\ 0 & 0 & 0 & k_1 & k_2 & 0 & k_3 & k_4 & 0 \\ 0 & 0 & 0 & 0 & k_1 & k_2 & 0 & k_3 & k_4 \end{pmatrix}}_{K'_{conv}} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_9 \end{pmatrix}$$

$$\mathrm{vec}(y) = K_{conv} \cdot \mathrm{vec}(x)$$

$$\nabla_k L = X * \nabla_y L$$

$$y_{ij} = \sum_{p,q} k_{pq} \, x_{i+p,\, j+q} \qquad \nabla_x L = \nabla_y L * \mathrm{reshape}(k)$$

$$dL = \sum_{i,j} \frac{\partial L}{\partial y_{ij}} \, dy_{ij} = \sum_{i,j} \frac{\partial L}{\partial y_{ij}} \sum_{p,q} \left(dk_{pq} \cdot x_{i+p,\, j+q} + \right.$$

$$\left. + k_{pq} \, dx_{i+p,\, j+q} \right) = \sum_{p,q} dk_{pq} \sum_{i,j} \frac{\partial L}{\partial y_{ij}} x_{i+p,\, j+q} +$$

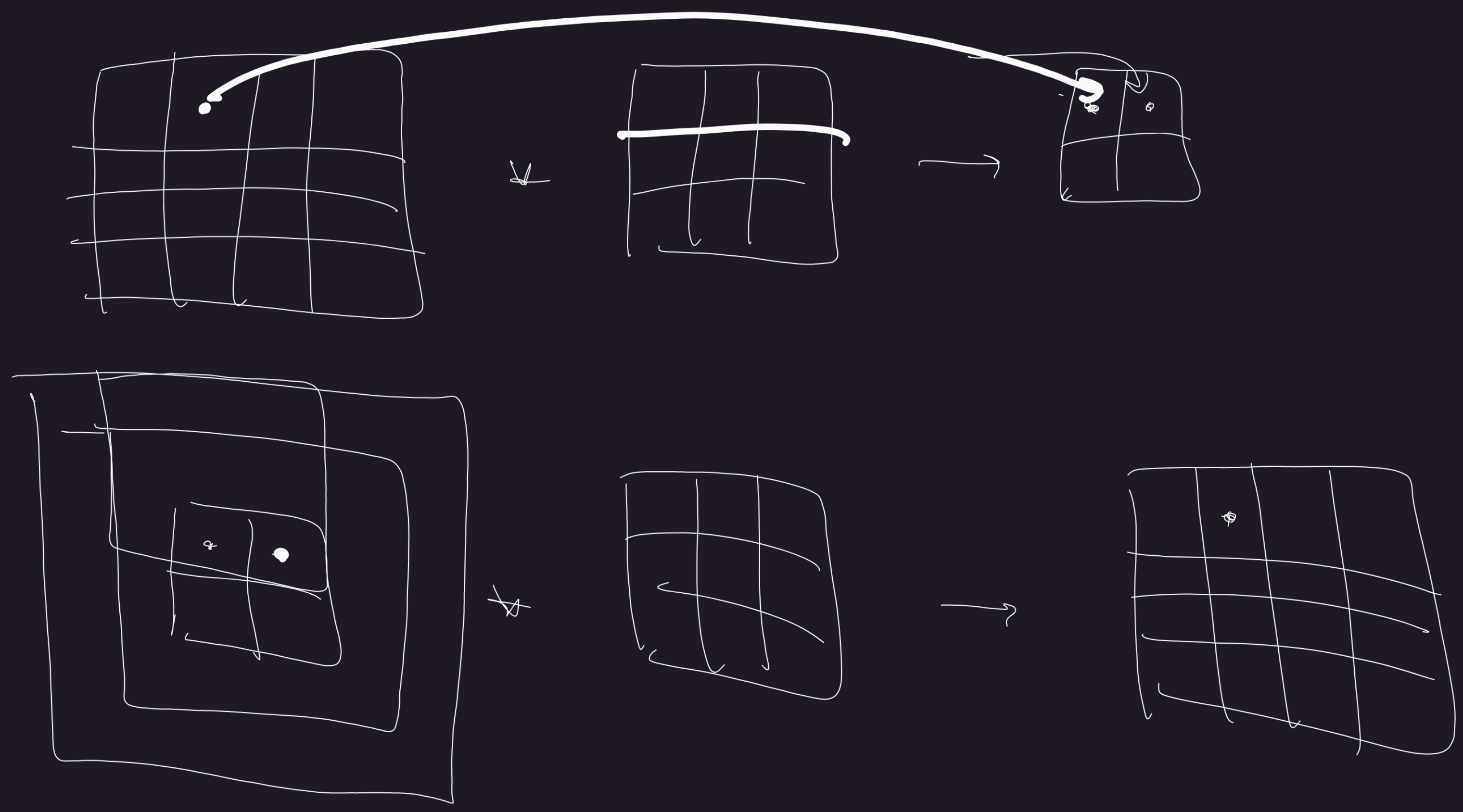
$$+ \sum_{u,v} dx_{u,v} \sum_{p,q} k_{pq} \frac{\partial L}{\partial y_{u-p,\, v-q}}$$

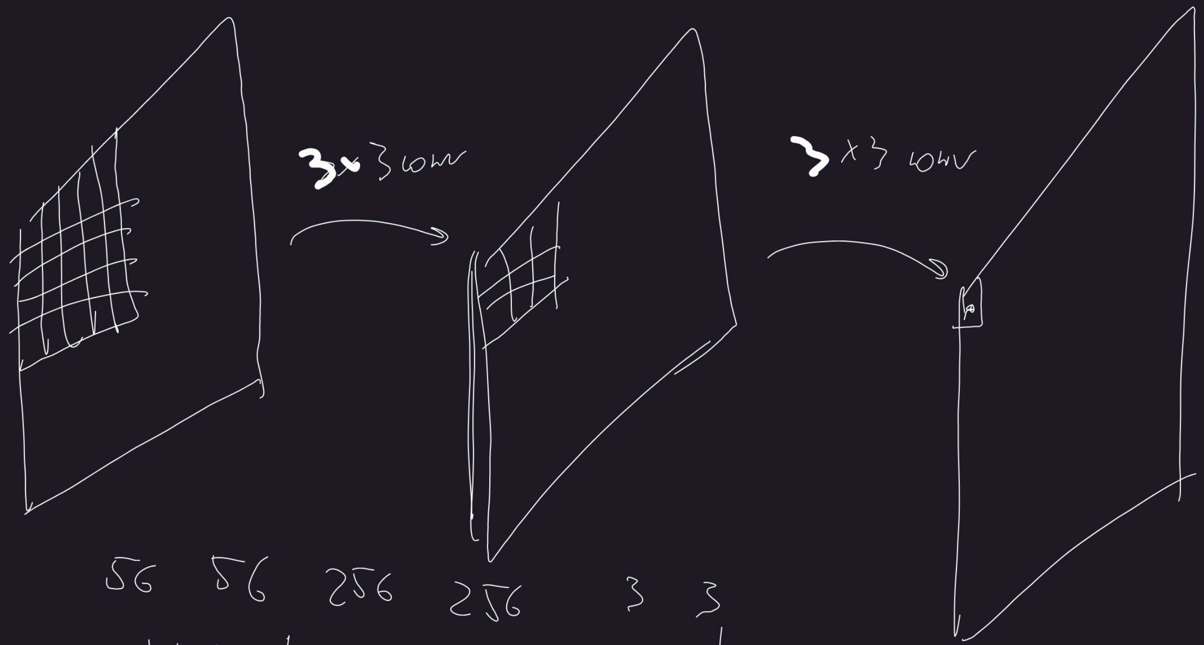
$$\sum_{i,j} \frac{\partial L}{\partial y_{ij}} x_{i+p,\, j+q} = \frac{\partial L}{\partial k_{pq}}$$

$$u = i+p \qquad v = j+q$$

$$\frac{\partial L}{\partial x_{uv}} = \sum_{p,q} k_{pq} \frac{\partial L}{\partial y_{u-p,\, v-q}}$$

	Linear	Conv2d
Forward	$z = Wx + b$	$z = X * K + b$
Backward	$\nabla_x L = W^T \nabla_z L$ $\nabla_W L = \nabla_z L \cdot x^T$ $\nabla_b L = \nabla_z L$	$\nabla_K L = X * \nabla_z L$ $\nabla_X L = \nabla_z L * \text{reshape}(K)$ $\nabla_b L = \text{sum}(\nabla_z L)$
# params	$n_{inputs} \cdot n_{outputs} + n_{outputs}$	$w_K \cdot h_K \cdot n_{input\_channels} \cdot n_{output\_channels} + n_{output\_channels}$
Comp. complexity	$O(n_{batch} \cdot n_{inputs} \cdot n_{outputs})$	$O(n_{batch} \cdot w_K \cdot h_K \cdot n_{input\_channels} \cdot n_{output\_ch})$
input size	(n_{batch}, n_{inputs})	$(n_{batch}, n_{in\_channels}, W, H)$
output size	(n_{batch}, n_{out})	$(n_{batch}, n_{out\_ch}, W, H)$





$5^2 = 25$

$3^2 + 3^2 = 18$

56 56 256 256 3 3

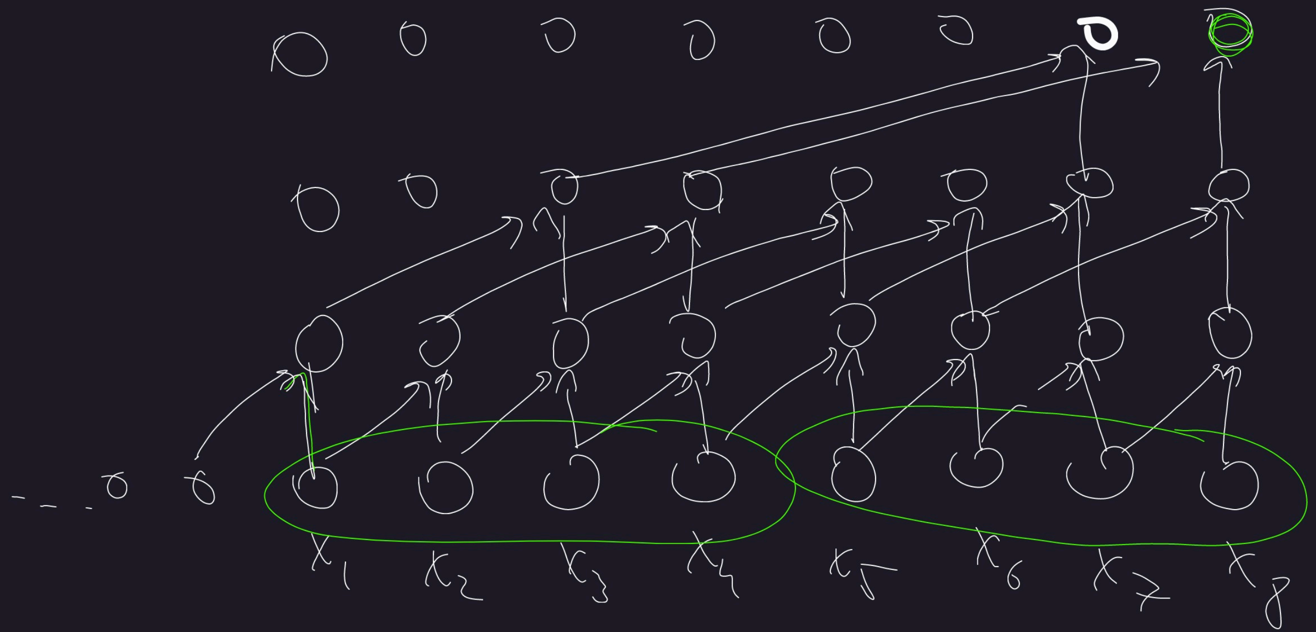
$H \cdot W \cdot n_{in} \cdot n_{out} \cdot w_k \cdot h_k$

$28 \to \frac{H}{2} \cdot \frac{W}{2} \cdot (n_{in} \cdot 2)(n_{out} \cdot 2) \, w_k \cdot h_3$

28 28 512 512 3 3


7
AvgPool
512
7
512

Temporal Convolutional NN



conv1d, kernel=2, dil=4

conv1d, kernel=2, dil=2

conv1d, kernel=2